

Name: _____

ID: _____ Section: _____

Instructions:

- Answer all questions.
- Marks are shown in brackets, and difficulty is shown as a tag (e.g., CA-E, CA-M, CA-D).
- Understanding the question is part of the exam.
- Write the answers for Questions 1(a), 1(b), 1(e), 2(a), 3(a), 3(b), and 3(c) in the designated answer boxes in this question paper.
- Write the answers for Questions 1(c), 1(d), 2(b), 2(c), 2(d), 2(e), and 2(f) in the answer script.
- For conceptual questions, write only the precise and relevant answer in short. Do not write detailed or descriptive explanations.

1. CO2 File Systems

[12]

- a) Suppose a directory /CSE321 which contains files and subdirectories. Safin navigates from /CSE321/lectureslides to its parent directory using “cd..”.

Explain how directory entries are stored and how it enables this navigation operation in the file system?

CA-E [2]

Answer:

- b) Analyze the journal entry given below.

CA-M [2]

TxB id=1	I(Q1) ptr: 10	D(Q1) address : 10	TxE id=1	TxB id=2	I(Q2) ptr: 20	D(Q2) address : 20	TxE id=2	TxB id=3	I(Q3) ptr: 10	TxE id=3
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- (i) **Identify** what type of issue will occur during the recovery after the system crashes? **[1]**

- (ii) **Mention** how the file system can prevent it from happening. **[1]**

Answer:

I	
II	

- c) A file system uses the UNIX inode data structure, which contains 7 direct block addresses, 1 single indirect block, 3 double indirect blocks, and 2 triple indirect blocks. The size of each block is 16 Bytes, and the size of each block address is 4 Bytes. Let's assume a reader is trying to read the 2880th Byte of the data from this file.

Find out how many disk read operations are needed to access that Byte? **CA-D [3]**

- d) A file system has an inode size = 512 Bytes and a block size = 2 KB. The first 5 blocks contain the superblock, journal, group descriptor, data bitmap, and inode bitmap. **CA-E [3]**

(i) **Calculate** the address of the inode number 14. **[1]**

(ii) The disk is sector addressable, and the size of each sector is 2 KB. **Calculate** the sector address of the inode block. **[2]**

- e) A file system creates and updates files very frequently. **Write** what will be the impact on file system performance if the space allocated for journal blocks is small? **CA-D [2]**

Answer:

2. CO3 Memory Management

[18]

- a) **Explain** how paging solves the problem of external fragmentation. **CA-E [2]**

Answer:

- b) Consider an operating system with a 32-bit logical address, and a page size of 4KB, running on 4GB RAM. The page table for each process is indexed with the page number, and contains the frame index, three bits for permission and three additional bits. Assuming a process uses up its maximum available memory in the operating system, **calculate** the size of the page table of the process. **CA-M [3]**

- c) Process A fork Process B, and both execute the following code snippet. Assume that the array arr is initially not loaded in RAM. The page size is 4KB.

Calculate what will be the i) maximum and ii) minimum number of pages newly created after the completion of the code snippet? **CA-D [1.5+1.5=3]**

```
1 for (int i = 0; i < 1200; i++) {  
2     arr[i] = arr[i] * 2  
3 }
```

- d) Consider a system with TLB miss ratio of 2.35%, and the main memory access time of 230 ns. If Effective Access Time is 247.405 ns, then **calculate** the TLB access time. **CA-E [3]**

- e) The OS is attempting to create a process that utilizes a single-level paging system. The current Free Frame List has no two consecutive free frames in the RAM, but there are numerous free frames available. The page size is 1KB. And the logical address space is 20 bits. **Write** can the process be created? Provide necessary mathematical calculations. **CA-D [3]**

- f) A process runs in a system with 3-level paging, and it has a logical address space of 16 bits. In the system, the page size is 16 Bytes, the size of each entry in the page table is 1 Byte, and the size of main memory is 1 KB.

After paging, the logical address space of the process looks like:

CA-M [2*2=4]

2nd outer pg#	outer pg#	inner pg#	offset
P1	P2	P3	d
4	4	4	4

Necessary page table information is given below:

2nd outer PMT		
Page #	Frame #	V/I bit
0		
1		
2		
3		
4		
5		
6		
7	12	v
8		
9		
10		
11		
12	24	v
13		
14		
15		

outer PMT						inner PMT					
Frame 12			Frame 24			Frame 18			Frame 5		
Pg#	Fr#	V/I bit	Pg#	Fr#	V/I bit	Pg#	Fr#	V/I bit	Pg#	Fr#	V/I bit
0			0			0			0		
1			1			1			1		
2			2			2			2	4	v
3			3			3	60	v	3		
4			4			4			4		
5	7	v	5			5			5		
6			6			6			6		
7			7	18	v	7			7		
8			8			8			8		
9			9			9			9		
10			10			10			10		
11			11	99	v	11			11	71	v
12			12			12			12		
13			13			13			13		
14			14			14	7	v	14		
15	5	v	15			15			15		

In this system, if the CPU generates logical addresses i) 51005 and ii) 32689, then **map** the corresponding physical addresses of these logical addresses. If the physical address is not valid, then write “invalid”.

3. CO1 Protection and Security

[5]

- a) Your OS enforces that whenever a domain is created, it is static and its contents cannot be changed. But you have to add some additional objects to your access list for a specific task after the domain has been initialized for it.

Identify what can be the possible solution to this problem?

CA-M [2]

Answer:

b) Consider the code snippet given below:

CA-M [2]

```
1  #include <stdlib.h>
2  #include <string.h>
3
4  struct log_entry {
5      char title[8];
6      char *msg;
7  };
8
9  void log_event(char *title, char *msg) {
10     struct log_entry *entry = malloc(sizeof(log_entry));
11     entry->msg = malloc(256);
12
13     strcpy(entry->title, title);
14     strncpy(entry->msg, msg, 256);
15
16     add_to_log(entry);
17 }
```

(i) **Identify** Which line of the code is responsible for the memory safety vulnerability? [1]

(ii) **Justify** the reason behind this. [1]

Answer:

I	
II	

c) A user successfully logs into an online banking system. While the session is still active, an attacker captures the session ID from an unsecured network and uses it to access the bank account without entering a username or password.

Identify the security violation method used in this scenario.

CA-E [1]

Answer:

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